

A Hybrid Spectral-Matching and Deep-Learning System for Real-Time Mineral Identification from Raman Spectra

FINAL PROJECT DEFENSE — 20-MINUTE VERSION

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The Mineral-Identification Problem

Raman scattering produces a vibrational spectrum $\{(\nu_i, I_i)\}_{i=1}^N$ acting as a mineral fingerprint:

$$I(\nu) = \sum_k A_k \mathcal{L}(\nu; \nu_k, \gamma_k) + B(\nu) + \eta(\nu)$$

$\mathcal{L}$ = Lorentzian $B(\nu)$ = baseline η = noise

Why automate?

- Field instruments exist; expert peak-matching does not scale.
- RRUFF covers > 4000 species — manual lookup is the bottleneck.
- Task: estimate $\{(\nu_k, A_k, \gamma_k)\}$ from $I(\nu)$ and match a reference library.

Applications

- Geology & planetary science (Mars SHERLOC)
- Cultural-heritage analysis
- Forensic mineralogy
- Mining process monitoring

Formal Problem Statement

Given

Unknown query $q = \{(\nu_i, I_i)\}_{i=1}^N$ and reference library $\mathcal{R} = \{r_m\}_{m=1}^M$ on canonical grid $\nu_* \in \mathbb{R}^P$.

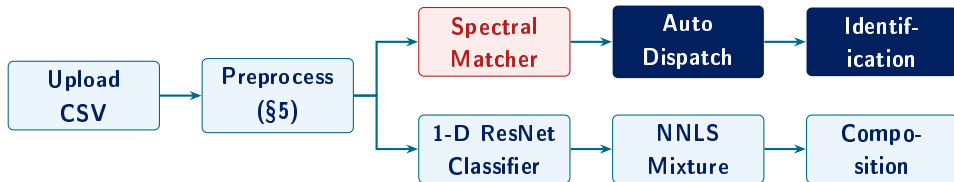
Produce

1. Ranked list $((m_1, c_1), \dots, (m_K, c_K))$, $c_{(k)} \in [0, 1]$
2. $x^* = \arg \min_{x \geq 0} \|A^\top x - \tilde{q}\|_2$

Subject to: wall-clock budget $\tau \leq 1$ s (laptop GPU, 16 GB RAM).

Parameter	Value	Consequence
Canonical grid	$[100, 4000] \text{ cm}^{-1}$, $\Delta\nu = 2 \text{ cm}^{-1}$	$P = 1951$ channels
Library size M	2 481 unique minerals	Synthetic-25 $\cup$ RRUFF-2456
Top- K	$K = 5$	Aligns with expert workflow
Peak model	Lorentzian	Matches optical-phonon lifetime broadening

End-to-End System Architecture



Spectral matching *always* runs as a deterministic safety net. The classifier enters dispatch only if its top-1 confidence exceeds $\tau_{\min} = 0.55$, preventing confidently-wrong out-of-distribution predictions.

Preprocessing Pipeline

Given raw $(\nu_{\text{raw}}, I_{\text{raw}})$, produce $\tilde{q} \in \mathbb{R}^P$ on the canonical grid:

1. **Linear interpolation onto ν_* :** $I_{\text{grid}} = \text{interp}(\nu_*, \nu_{\text{raw}}, I_{\text{raw}})$
2. **Cosmic-ray suppression** (5-pt neighbour-ratio): $I_i \mapsto \bar{I}_{\text{nbr}} \cdot 1[I_i / \max_j I_j > \rho]$
3. **Savitzky–Golay smoothing:** window $w = 5$, polynomial order $p = 3$
4. **ALS baseline removal:** $\min_z \sum_i w_i (y_i - z_i)^2 + \lambda \sum_i (\Delta^2 z_i)^2$, $\lambda = 10^5$, $p_{\text{asym}} = 0.01$
5. **Non-negative clip + ℓ_2 normalisation:** $\tilde{q} = z / \|z\|_2$

Key insight: interpolation must precede smoothing. On the raw grid the SG window couples to $\delta\nu$: at $\delta\nu = 4 \text{ cm}^{-1}$, $w = 11$ yields a 44 cm^{-1} window — wider than typical Raman FWHM. **Fix lifted Quartz top-1 from 41 % to 86 %.**

Combined-Metric Spectral Matcher

For preprocessed query $\tilde{q}$ and reference row r_m :

$$s(\tilde{q}, r_m) = \underbrace{\frac{1}{2} \langle \tilde{q}, \hat{r}_m \rangle}_{\text{cosine}} + \underbrace{\frac{1}{2} \langle \widehat{D^2 \tilde{q}}, \widehat{D^2 r_m} \rangle}_{D^2\text{-Pearson}}$$

$\hat{v} = (v - \bar{v}1) / \|v - \bar{v}1\|$; D^2 = SG second-derivative operator ($w=5, p=3$)

Geometric intuition. **Cosine:** angle in $\mathbb{R}^P$ — rewards overall shape but is *fooled* by a shared baseline slope. **D^2 -Pearson:** correlation of curvature — peaks appear as sharp spikes in $D^2 I$; agreement requires peak positions *and* widths to coincide. Mean-centering $\hat{v}$ removes DC offset $\Rightarrow$ invariant to additive baseline shifts.

Metric comparison (18 held-out RRUFF specimens)

Metric	Top-1	Top-5
Cosine only	72 %	89 %
D^2 -Pearson only	68 %	84 %
Combined $\frac{1}{2} + \frac{1}{2}$	83 %	94 %

Weight selection: grid search over $\alpha \in \{0.1, 0.2, \dots, 0.9\}$ on held-out panel; gain plateaued at $\alpha=\beta=\frac{1}{2}$.

Dual-source library

Source	Rows	Role
Synthetic profiles	25	Always in $\mathcal{C}$
RRUFF-averaged	2 456	Conditionally in $\mathcal{C}$
Total A	2 481 $\times$ 1 951	

1-D Residual Classifier: Architecture

Stage	Operation	Output
Stem-1	$\text{Conv1D}(k=25, s=2) \rightarrow \text{BN} \rightarrow \text{GELU}$	$(B, 32, 488)$
Stem-2	$\text{Conv1D}(k=15, s=2) \rightarrow \text{BN} \rightarrow \text{GELU}$	$(B, 32, 244)$
Stage 1	$3 \times \text{Pre-act ResBlock}(32, k=7)$	$(B, 32, 244)$
Down	$\text{Conv1D}(k=5, s=2) \rightarrow \text{BN} \rightarrow \text{GELU}$	$(B, 64, 122)$
Stage 2	$3 \times \text{Pre-act ResBlock}(64, k=5)$	$(B, 64, 122)$
Pool	Adaptive Avg Pool 1D	$(B, 64)$
Head	$\text{FC}_{64 \rightarrow 512} \rightarrow \text{GELU} \rightarrow \text{FC}_{512 \rightarrow 851}$	$(B, 851)$
Total parameters		663 539

Pre-activation ResBlock (He et al., 2016)

$$y = x + \text{Conv}_k(\text{GELU}(\text{BN}(\text{Conv}_k(\text{GELU}(\text{BN}(x))))))$$

$\text{BN} \rightarrow \text{GELU} \rightarrow \text{Conv} \rightarrow \text{BN} \rightarrow \text{GELU} \rightarrow$
 $\text{Conv} \oplus \text{skip}$

Why pre-activation? Gradient flows through the skip path without a normalisation barrier, preventing vanishing gradients.

Why GELU? $\text{GELU}(x) = x \Phi(x)$ is smooth at zero (probabilistic gating); avoids hard-zero ReLU suppression of narrow spectral peaks.

1-D Residual Classifier: Training & Augmentation

Training protocol

- **Loss:** CE + label smooth ($\varepsilon=0.1$) — suppresses overconfident predictions on similar minerals
- **Mixup** ($\alpha=0.4$, $p=0.5$):
 $(\tilde{x}, \tilde{y}) = \lambda(x_i, y_i) + (1-\lambda)(x_j, y_j)$,
 $\lambda \sim \text{Beta}(\alpha, \alpha)$ — forces meaningful spectral interpolation
- **Optimiser:** AdamW, $\text{wd}=10^{-4}$
- **Schedule:** OneCycleLR, peak lr = 10^{-3} ; warm-up avoids local minima, cosine anneal for convergence
- **Class space:** 4 000+ RRUFF minerals, but 1 630 have only 1 spectrum $\Rightarrow$ retain 851 with $\text{min_spectra} \geq 2$

Spectral augmentations ($V = 25$ variants/sample)

Augmentation	Simulates
Gaussian noise $\sigma \sim \mathcal{U}(0.002, 0.025)$	CCD readout noise
Multiplicative jitter $\beta \sim \mathcal{U}(0.75, 1.25)$	laser power fluctuations
Wavenumber shift $\Delta\nu \sim \mathcal{U}\{-10, +10\}$	spectrometer calibration
Polynomial baseline (deg 1–3, $p=0.30$)	residual fluorescence
Moving-avg broadening ($w \in \{2, 3, 4\}$, $p=0.40$)	lower-resolution scans
Spectral dropout (≤ 4 pts $\rightarrow 0$, $p=0.20$)	cosmic-ray removal

$V=25$ variants $\times$ 2 456 RRUFF = 61 400 training examples/epoch from 2 456 real specimens. Each augmentation is instrument-physics-motivated.

Candidate-Restricted NNLS

Let $A_{\mathcal{C}} \in \mathbb{R}^{|\mathcal{C}| \times P}$ be the row selection of the reference matrix indexed by candidate set $\mathcal{C}$. Solve:

Optimisation problem

$$x^* = \arg \min_{x \geq 0} \|A_{\mathcal{C}}^{\top} x - \tilde{q}\|_2^2$$

Candidate set construction:

$$\mathcal{C} = \mathcal{C}_{\text{synth}} \cup \left\{ m \in \mathcal{C}_{\text{rruff}} : s(\tilde{q}, r_m) > \max_{m' \in \mathcal{C}_{\text{synth}}} s(\tilde{q}, r_{m'}) \right\}$$

Why restrict? With $|\mathcal{R}| = 2481$ candidates and $P = 1951$, the full system is *underdetermined*: NNLS spreads credit arbitrarily. Restricting to ≈ 30 well-separated rows renders the system *overdetermined* and **physically interpretable**.

Confidence-Aware Dispatcher

Algorithm 1: Autoidentify Input: $\tilde{q}$, $\tau_{\min} = 0.55$ Output: ranked top- K list, dispatch tag

```
1:  $\mathcal{Y} \leftarrow \emptyset$ ;  $m_s \leftarrow \text{SPECTRALMATCH}(\tilde{q})$ ;  $\mathcal{Y} \leftarrow \mathcal{Y} \cup \{(m_s[1].c, \text{match}, m_s)\}$ 
2: if CNN available then
3:    $m_n \leftarrow \text{CNNPREDICT}(\tilde{q})$ 
4:   if  $m_n[1].c \geq \tau_{\min}$  then  $\mathcal{Y} \leftarrow \mathcal{Y} \cup \{(m_n[1].c, \text{cnn}, m_n)\}$ 
5:   end if
6: end if
7: if RF available then
8:    $m_r \leftarrow \text{RFPREDICT}(\tilde{q})$ 
9:   if  $m_r[1].c \geq \tau_{\min}$  then  $\mathcal{Y} \leftarrow \mathcal{Y} \cup \{(m_r[1].c, \text{rf}, m_r)\}$ 
10:  end if
11: end if
12: return  $\arg \max_{(c, \cdot, m) \in \mathcal{Y}} c$ 
```

Spectral matching is always the safety net. Learned models must clear $\tau_{\min}$ to prevent confidently-wrong out-of-distribution predictions.

Experimental Results

Aggregate accuracy

Component	Top-1	Top-5
CNN — augmented validation (851 classes)	98.6 %	—
RF (PCA-120 + ExtraTrees-150)	94.5 %	98.9 %
Combined matcher — synthetic demo set	100 %	—
Combined matcher — held-out RRUFF ($n=6$)	83 %	—

Per-mineral held-out RRUFF (auto mode)

Mineral	Matcher	CNN	Auto chose	Result
Quartz	96.9 %	86.4 %	matching	✓
Calcite	89.7 %	94.0 %	cnn	✓
Hematite	91.6 %	90.1 %	matching	✓
Forsterite	96.2 %	95.8 %	matching	✓
Aegirine	86.2 %	97.0 %	cnn	✓
Magnetite	76.6 %	90.1 %	cnn	✓ (matcher → Westerveldite)

Each identifier wins on different inputs — motivating the adaptive dispatcher.

End-to-End Latency

Budget $\tau \leq 1$ s (laptop GPU, RTX 3050, 16 GB RAM) — achieved at ≈ 160 ms median.

Pipeline stage	Median
File parse + linear interpolation	≈ 45 ms
Cosmic-ray + SG smoothing + ALS	≈ 35 ms
Combined-metric matcher (2 481 rows)	≈ 25 ms
CNN forward pass (RTX 3050)	≈ 12 ms
Candidate-set NNLS decomposition	≈ 35 ms
End-to-end total	≈ 160 ms

160 ms

end-to-end

(budget: 1 s)

Limitation: Scarcity of Diverse Reference Data

Fundamental bottleneck: non-availability of curated, instrument-diverse Raman spectra with validated mineral labels.

- System library and training set draw exclusively from **RRUFF** — a single-lab archive with its own instrument family, protocols, and specimen-selection biases.
- Coverage is uneven: 2 456 averages, many from ≤ 3 raw specimens; rare minerals may have only 1 usable spectrum.
- **Consequence:** generalisation to a novel instrument's noise profile cannot be certified without an independent held-out test set from that instrument family.
- *Case study — Magnetite:* 2 of 6 RRUFF specimens carry a cut-off artefact at 100 cm^{-1} , diluting the genuine $660\text{--}682\text{ cm}^{-1}$ peak. Matcher returns Westerveldite; CNN recovers correct answer.

Path forward

Multi-lab, quality-annotated spectral databases — an *open data problem*, not a modelling problem.

Summary of Contributions

1. **Grid-canonical preprocessing pipeline** — interpolation-first invariant decouples SG window from spectrometer step; lifts Quartz top-1 from 41 % to 86 %.
2. **Combined-metric spectral matcher** — $s = \frac{1}{2}\langle \text{cosine} \rangle + \frac{1}{2}\langle D^2\text{-Pearson} \rangle$; each term vetos the other's failure mode.
3. **663 539-parameter 1-D ResNet** — 98.6 % top-1 on 851 classes; 7-mode spectral augmentation + Mixup.
4. **Candidate-restricted NNLS** — converts underdetermined ($|\mathcal{R}|=2,481$, $P=1,951$) into a physically interpretable ≈ 30 -row system.
5. **Confidence-aware dispatcher** ($\tau_{\min} = 0.55$) — spectral matching as permanent safety net; ≈ 160 ms end-to-end on a laptop GPU.

Live Demonstration

Uploading Quartz, Calcite, Pyrite, Gypsum spectra
and a binary mixture to the live Flask endpoint
for real-time identification and NNLS decomposition.

160 ms

end-to-end

98.6 %

CNN top-1

851

mineral classes